Team Number:
Project Name:
Team Member Names:
Version:

A. List ALL major components (a
All Major Components

B. Assign each major component
+5V Power Rail

c1. Regulator or Source Choice

C. For each power rail above, sel

D. Select a specific external powe
External Power Source 1
Power Source 1 Selection
Power Rails Connected to External Power Source 1

E. Calculate Battery Life (if applic

Notes

External Supply Voltage should be

If you have multiple units in your de

Branden Cluff'

202
Weather System
Zander Boward, Sidra Elsaady, Brendan Laskowski
1

ctive devices, integrated circuits, etc.) except for power sources, voltage regulato

Component Name	Part Number
PIC18F57Q43 Curiosity Nano	PIC18F57Q43
MCP6004 Quad Op-Amp	MCP6004
PDV-P8103 Photoresistor	PDV-P8103

t above to ONE power rail below. Try to minimize the number of different power ra

Component Name	Part Number
PIC18F57Q43 Curiosity Nano	PIC18F57Q43
MCP6004 Quad Op-Amp	MCP6004
PDV-P8103 Photoresistor	PDV-P8103

STMicroelectronics L7805CV Linear Regulator	L7805CV

ect a specific voltage regulator using the same process as for major component s

er source (wall supply or battery) for your system, and confirm that it can supply a

Component Name	Part Number
9V DC Wall Adapter	WSU090-1300-13
+5V Regulator STMicroelectronics L7805CV Linear Regulator	LM7805CV

cable). For each battery, also check the worst-case lifetime of the battery by indic

Component Name	Part Number
Standard 9V Alkaline Battery	6LR61-1B

determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V r

sign (e.g., a base unit and remote unit) then you need a separate power budget for eacl

s Power Budget

ors, resistors, capacitors, or passive elements		
SupplyVoltageRange	**#**	**AbsoluteMaximumCurrent (mA)**
+3.3V - 5.0V	1	20
+1.8V - 5.5V	1	2
Variable/Passive	1	0.5

ails in the design.		
SupplyVoltageRange	**#**	**AbsoluteMaximumCurrent (mA)**
+3.3V - 5.0V	1	20
+1.8V - 5.5V	1	2
Variable/Passive	1	0.5

Subtotal

Safety Margin

Total Current Required on +5V Rail

+7V-35V	1	1000

Total Remaining Current Available on +12V Rail

Selection. Confirm that the Total Remaining Current Available on each rail abo

all of the regulators for all of the power rails simultaneously. If you need multi		
SupplyVoltageRange	**Output**	**AbsoluteMaximumCurrent (mA)**
+90 - 264 VAC	+9V	1300
(range)	1	1000

Total Remaining Current Available on External Power Source 1

ating the capacity in mAh.	
SupplyVoltageRange	**Capacity(mAh)**
+9V	480

Battery Life

egulator).

h unit

TotalCurrent(mA)	Unit
20	mA
2	mA
0.5	mA

TotalCurrent(mA)	Unit
20	mA
2	mA
0.5	mA
0	mA
0	mA
22.5	mA
25%	
28.125	mA
1000	mA
971.875	mA

ve is not negative.

ble power sources, list each

TotalCurrent(mA)	Unit
1300	mA
0	mA
1000	mA
0	mA
300	mA

RequiredByRegulators	
22.5	
21.33333333	hours